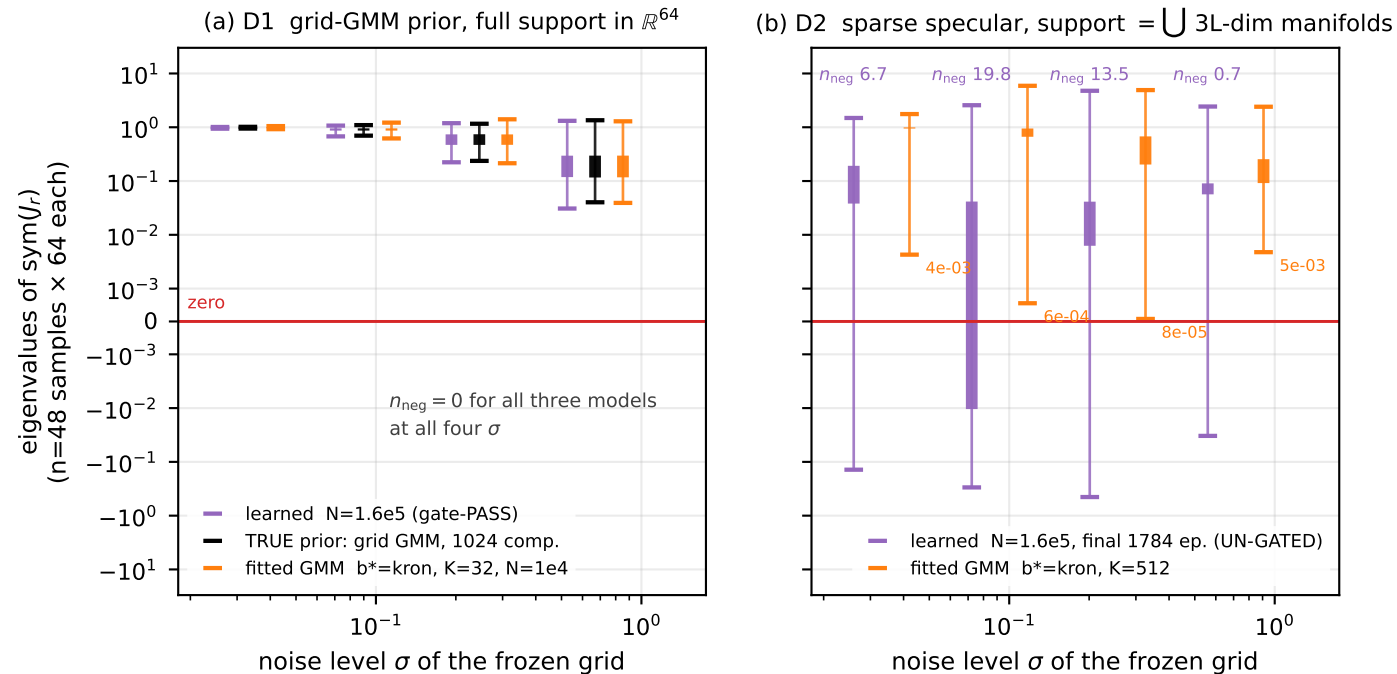


The EP matrix site (D-14) inverts $\nu \text{Herm}(J)$ and therefore needs J PSD. It is, on D1. It is not, on D2.



Whisker = $[\lambda_{\min}, \lambda_{\max}]$, bar = 25th percentile..median of the 48×64 eigenvalues at that σ . y is symlog (linear below 10^{-3}): the sign change and the near-zero bulk are both on the page, nothing is clipped.

On D2 at $N=1\text{e}4$ the same n_{neg} counts are 8.0 / 10.6 / 11.2 / 7.8 (logs/jacspec_D2_N10000.log): 16x the data moves them 8.0 $\rightarrow$ 6.7, 10.6 $\rightarrow$ 19.8, 11.2 $\rightarrow$ 13.5, 7.8 $\rightarrow$ 0.7 -- worse at two of four σ , not merely no better.

At $\sigma = 0.092$ the learned 25th percentile is -0.0104 : over a quarter of the 64 eigenvalues are below zero, which is why that bar starts under the red line. The fitted GMM on D2 stays POSITIVE but falls to 7.6×10^{-5} (per-point values beside its lower whisker, rounded to one significant figure): near the PSD boundary, not on it.